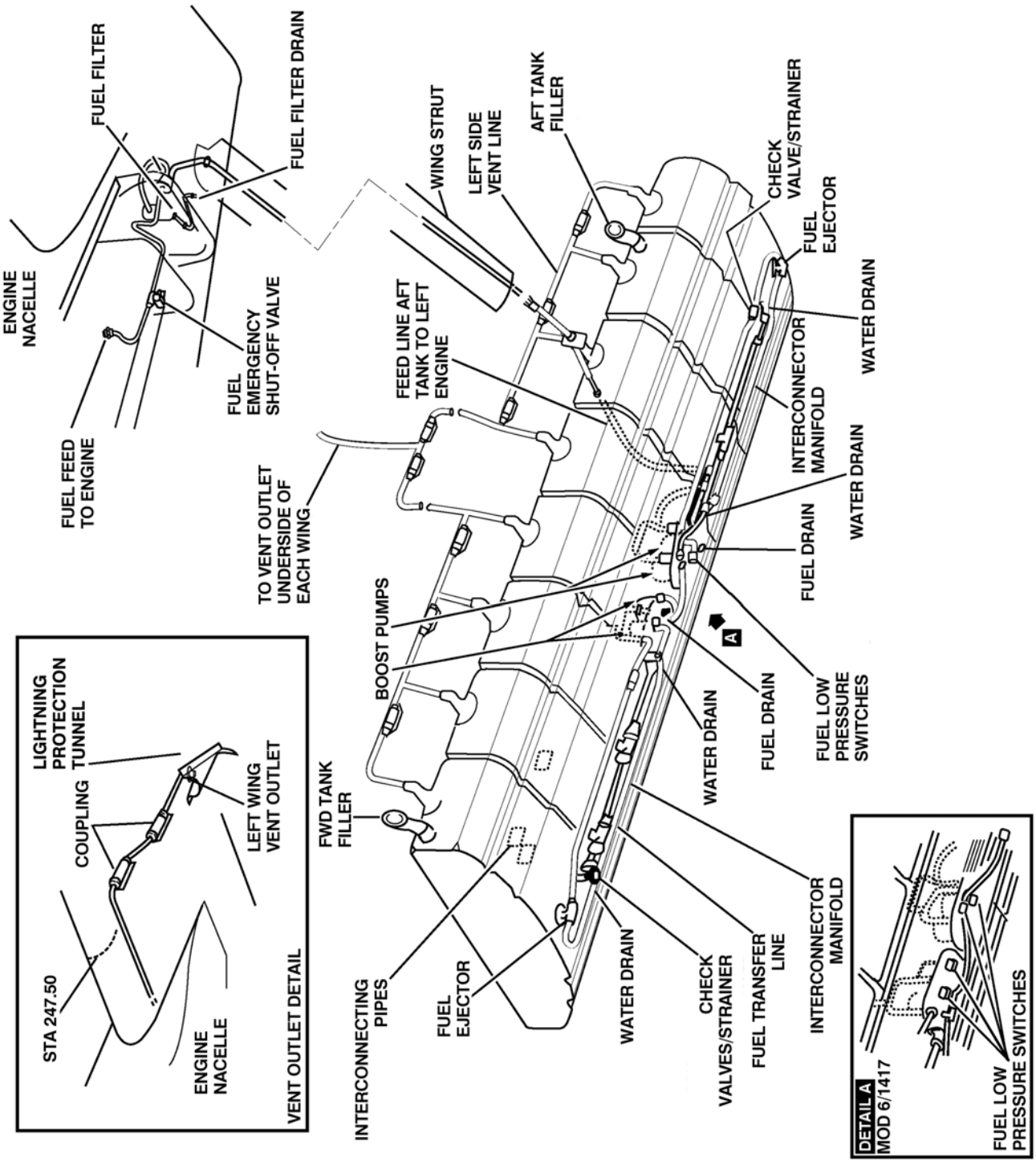
Fuel (Mod 6/2006) – General Data

1. General

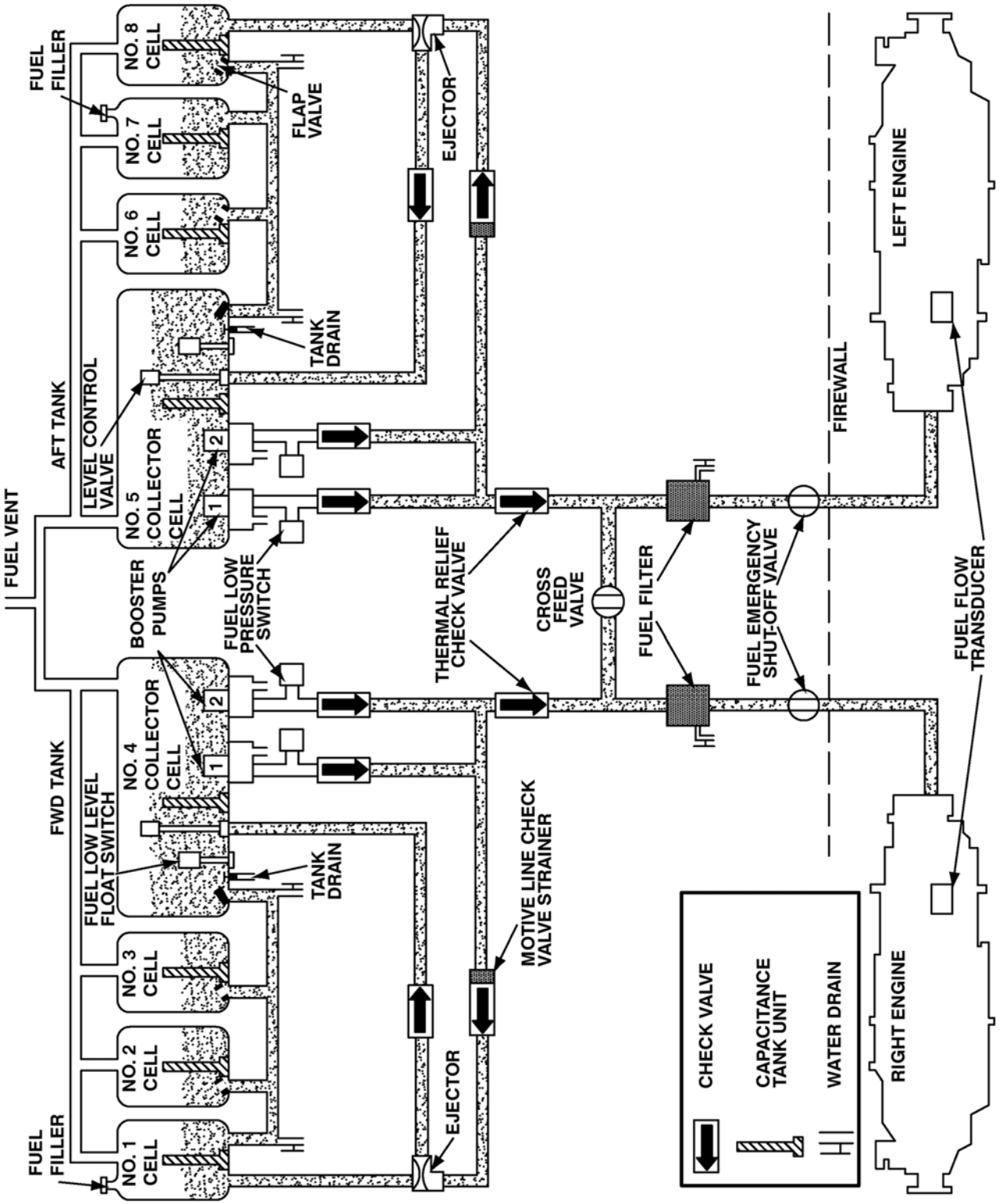
Refer to Figure 1 and Figure 2 for details.

Fuel is contained in two fuselage fuel tanks, one forward and one aft. The tanks can be gravity refueled through two filler points, one for each tank, located on the left side of the fuselage near the wing strut. The forward tank supplies the right engine and the aft tank the left engine, with the main feed lines to both engines passing through No. 6 cell (aft tank). A crossfeed valve provides a fuel supply to both engines from one tank. Each fuel tank consists of four interconnected fuel cells. Two fuel booster pumps, installed in the collector cell of each tank, deliver fuel under pressure to the engine; this pressure also operates a fuel ejector which transfers fuel from the end cell into the collector cell. From No. 6 cell the main feed line to each engine is routed up the rear side of the wing strut, along the wing and into the appropriate nacelle. Mounted in each nacelle behind the firewall is a fuel filter and a fuel emergency shut-off valve. To permit checks for water accumulation in the fuel system, two water drain valves are provided in each of the forward and aft fuselage tank interconnector manifolds and in each fuel filter. In addition, a fuselage tank fuel drain valve is provided in each sump plate for No. 4 and No. 5 collector cells. A fuel quantity indicating system of the capacitance type is provided for each fuel tank. Fuel low pressure and low level are indicated through Apex on the fuel system window; the upper right 1/6th of the lower Multi-Function Display (MFD). Refer to the Wiring Diagram Manual PSM 1-64-2W for electrical schematic.



Fuel System Arrangement
Figure 1

Print Date: 2025-09-15



Fuel System Schematic
Figure 2

Print Date: 2025-09-15

2. Adjustment/Test

Note Aircraft must be reasonably level, and in the open air, during all fuel system checks.

A. Electrical Check (No Fuel)

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Ensure that fuel tanks are drained of all usable fuel and, with all relevant circuit breakers engaged, set switches in modes 1 to 5 given in Table 1 — Fuel System Electrical Check – No Fuel. Check that all conditions are produced as shown.

B. Transfer and Amber Block Message Check

1) On the fuel control panel, verify FWD BOOST PUMP 1 switch is in the OFF position and FWD BOOST PUMP 2 switch in the NORMAL position.

2) On the fuel control panel, verify AFT BOOST PUMP 1 switch is in the OFF position and AFT BOOST PUMP 2 switch is in the NORMAL position.

3) Place 30 Imperial (36 U.S.) gallons of fuel in forward fuselage tank and allow fuel level to stabilize for 5 minutes.

4) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.), engage relevant circuit breakers and observe that the forward boost pump amber blocks Pmp1 and Pmp2 are displayed and the Fwd Fuel Qty Low caution message is displayed.

Note Fuel in collector cell is below operating level of low level float switch.

5) On fuel control panel, set FUEL SELECTOR rotary switch to NORMAL and select forward fuselage tank BOOST PUMP 1 switch to the ON position.

6) Check that both forward boost pump amber block Pmp1 and Pmp2 messages are no longer displayed and Fwd Fuel Qty Low caution message is no longer displayed, within 3 minutes.

Note Fuel transfer takes place, raising fuel level in collector cell to operate low level float switch.

7) Switch off booster pump and verify Fwd Fuel Qty Low caution message is no longer displayed.

8) Repeat Para 2.B.3) through Para 2.B.7), substituting aft fuselage tank for forward.

C. Pressure Check

1) Ensure that each fuselage tank contains a minimum of 30 Imperial (36 U.S.) gallons of fuel.

2) Apply DC Power (refer to 12-00-00, Servicing – General Data, Para 2.) and engage relevant circuit breakers.

3) Position fuel levers in overhead console to OFF and FUEL SELECTOR rotary switch on fuel control panel to NORMAL. Verify that the fuel pump amber blocks Pmp1 and Pmp2 for the forward and aft fuel tank are displayed.

4) On fuel control panel, select FWD and AFT BOOST PUMP 1 switches to ON. Verify that the fuel pump amber blocks Pmp1 and Pmp2 for the forward and aft tank on the fuel pane are no longer displayed.

5) Inspect complete fuselage fuel system for leaks. Repair leaks before proceeding with pressure check.

6) Select and hold booster pump switches to TEST. Check that both No. 1 boost pump Pmp1 amber block messages are displayed and both No. 2 boost pump Pmp2 amber block messages are not displayed.

7) Release booster pump switches to OFF and check that all four fuel pump Pmp1 and Pmp2 amber block messages are displayed.

D. Function Check

1) Disconnect each engine fuel supply flexible hose, routed between emergency fuel shut-off valve on firewall and engine oil-to-fuel heater at elbow on heater. Place disconnected end of hose in a suitable container to receive fuel.

2) Check that forward fuselage tank contains approximately 30 Imperial (36 U.S.) gallons of fuel.

3) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.), engage relevant circuit breakers and, using Table 2 — Fuel System Function Check – Fuel In Forward Tank, check fuel system in modes 1 to 5.

4) With switches in mode 2 and fuel flow through engine supply flexible hose stabilized, measure and record time taken for one Imperial (1.2 U.S.) gallon of fuel to be pumped from forward fuselage tank. Maximum time allowed is 30 seconds.

5) With switches in mode 5 and fuel flow through engine supply flexible hose stabilized, measure and record time taken for one Imperial (1.2 U.S.) gallon of fuel to be pumped from forward fuselage tank. Maximum time allowed is 30 seconds.

6) Drain forward fuselage tank and check that aft fuselage tank contains approximately 30 Imperial (36 U.S.) gallons of fuel.

7) Repeat Para 2.D.3) through Para 2.D.5) for aft fuselage tank, substituting AFT for FWD and Table 3 — Fuel System Function Check – Fuel in Aft Tank for Table 2 — Fuel System Function Check – Fuel In Forward Tank.

8) Ensure that all boost pump switches are off and remove external power.

9) Reconnect engine fuel supply flexible hoses to engine oil-to-fuel heaters.

CAUTION **DO NOT ALLOW PUMPS TO RUN MORE THAN 30 SECONDS BEFORE RETURNING THE SWITCHES TO THE OFF POSITION.**

Table 1 Fuel System Electrical Check – No Fuel

SWITCHES	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5
Fuel Selector	NORMAL	BOTH ON FWD	BOTH ON AFT	NORMAL	NORMAL
Fuel Emergency Shut-off – Left	NORMAL	FUEL OFF	FUEL OFF	FUEL OFF	FUEL OFF
Fuel Emergency Shut-off – Right	NORMAL	FUEL OFF	FUEL OFF	FUEL OFF	FUEL OFF
Fwd Boost Pump 1	OFF	ON	ON	TEST	OFF
Aft Boost Pump 1	OFF	ON	ON	TEST	OFF
Left and Right Boost Pump 2	NORMAL	NORMAL	NORMAL	NORMAL	EMERGENCY
PRODUCED CONDITION	↓	↓	↓	↓	↓

Table 1 Fuel System Electrical Check – No Fuel (continued)

SWITCHES	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5
No. 1 Boost Pump Fwd	OFF	ON	OFF	OFF	OFF
No. 2 Boost Pump Fwd	OFF	ON	OFF	ON	ON
No. 1 Boost Pump Aft	OFF	OFF	ON	OFF	OFF
No. 2 Boost Pump Aft	OFF	OFF	ON	ON	ON
Crossfeed Valve	CLOSED	OPEN	OPEN	CLOSED	CLOSED
Shut-off Valve – Left	OPEN	CLOSED	CLOSED	CLOSED	CLOSED
Shut-off Valve – Right	OPEN	CLOSED	CLOSED	CLOSED	CLOSED
FUEL PANE AMBER BLOCK MESSAGE STATUS					
Fwd Pmp1	ON	ON	ON	ON	ON
Fwd Pmp2	ON	ON	ON	ON	ON
Aft Pmp1	ON	ON	ON	ON	ON
Aft Pmp2	ON	ON	ON	ON	ON
CAUTION MESSAGE STATUS					
Boost Pump 1 Fwd Press	ON	ON	OFF	ON	ON
Boost Pump 2 Fwd Press	ON	ON	OFF	ON	ON
Boost Pump 1 Aft Press	ON	OFF	ON	ON	ON
Boost Pump 2 Aft Press	ON	OFF	ON	ON	ON
WARNING MESSAGE STATUS					
Fuel Low Both Tanks	ON	ON	ON	ON	ON

CAUTION DO NOT ALLOW PUMPS IN EMPTY TANKS TO RUN MORE THAN 30 SECONDS.

Table 2 Fuel System Function Check – Fuel In Forward Tank

	FORWARD TANK				
SWITCHES	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5
Fuel Selector	BOTH ON FWD	NORMAL	BOTH ON FWD	NORMAL	NORMAL
Fuel Emergency Shut-off – Left	FUEL OFF	NORMAL	NORMAL	FUEL OFF	NORMAL

Table 2 Fuel System Function Check – Fuel In Forward Tank (continued)

SWITCHES	FORWARD TANK				
	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5
Fuel Emergency Shut-off – Right	FUEL OFF	NORMAL	NORMAL	FUEL OFF	NORMAL
Fwd Boost Pump 1	OFF	ON	ON	TEST	OFF
Aft Boost Pump 1	ON	OFF	OFF	OFF	OFF
Left and Right Boost Pump 2	NORMAL	NORMAL	NORMAL	NORMAL	BOTH TO EMERGENCY
PRODUCED CONDITION	↓	↓	↓	↓	↓
Full Bore Flow At Left Engine	NO FLOW	NO FLOW	YES	NO FLOW	NO FLOW
Full Bore Flow At Right Engine	NO FLOW	YES	YES	NO FLOW	YES
Fuel Quantity Indication – Fwd	NO CHANGE	CHANGE	CHANGE	NO CHANGE	CHANGE
Fuel Quantity Indication – Aft	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE
FUEL PANE AMBER BLOCK MESSAGE STATUS					
Fwd Pmp1	OFF	OFF	OFF	ON	ON
Fwd Pmp2	OFF	OFF	OFF	OFF	OFF
Aft Pmp1	ON	ON	ON	ON	ON
Aft Pmp2	ON	ON	ON	ON	ON
CAUTION MESSAGE STATUS					
Boost Pump 1 Fwd Press	OFF	OFF	OFF	ON	ON
Boost Pump 2 Fwd Press	OFF	OFF	OFF	OFF	OFF
Boost Pump 1 Aft Press	OFF	ON	OFF	ON	ON
Boost Pump 2 Aft Press	OFF	ON	OFF	ON	ON
Fwd Fuel Qty Low	OFF	OFF	OFF	OFF	OFF
Aft Fuel Qty Low	ON	ON	ON	ON	ON

CAUTION **DO NOT ALLOW PUMPS IN EMPTY TANKS TO RUN MORE THAN 30 SECONDS.**

Table 3 Fuel System Function Check – Fuel in Aft Tank

SWITCHES	AFT TANK				
	MODE 1	MODE 2	MODE 3	MODE 4	MODE 5
Fuel Selector	BOTH ON AFT	NORMAL	BOTH ON AFT	NORMAL	NORMAL
Fuel Emergency Shut-off – Left	FUEL OFF	NORMAL	NORMAL	FUEL OFF	NORMAL
Fuel Emergency Shut-off – Right	FUEL OFF	NORMAL	NORMAL	FUEL OFF	NORMAL
Fwd Boost Pump 1	ON	OFF	OFF	OFF	OFF
Aft Boost Pump 1	OFF	ON	ON	TEST	OFF
Left and Right Boost Pump 2	NORMAL	NORMAL	NORMAL	NORMAL	BOTH ON EMERGENCY
PRODUCED CONDITION	↓	↓	↓	↓	↓
Full Bore Flow At Left Engine	NO FLOW	YES	YES	NO FLOW	YES
Full Bore Flow At Right Engine	NO FLOW	NO FLOW	YES	NO FLOW	NO FLOW
Fuel Quantity Indication – Fwd	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE	NO CHANGE
Fuel Quantity Indication – Aft	NO CHANGE	CHANGE	CHANGE	NO CHANGE	CHANGE
FUEL PANE AMBER BLOCK MESSAGE STATUS					
Fwd Pmp1	ON	ON	ON	ON	ON
Fwd Pmp2	ON	ON	ON	ON	ON
Aft Pmp1	OFF	OFF	OFF	ON	ON
Aft Pmp2	OFF	OFF	OFF	OFF	OFF
CAUTION MESSAGE STATUS					
Boost Pump 1 Fwd Press	OFF	ON	OFF	ON	ON
Boost Pump 2 Fwd Press	OFF	ON	OFF	ON	ON
Boost Pump 1 Aft Press	OFF	OFF	OFF	ON	ON
Boost Pump 2 Aft Press	OFF	OFF	OFF	OFF	OFF
Fwd Fuel Qty Low	ON	ON	ON	ON	ON
Aft Fuel Qty Low	OFF	OFF	OFF	OFF	OFF